

Algebra II with Trigonometry

Chapter 5.6

Mental Math (Gemini)

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[4401272_alg-2trig-h-chp-5-6-note-docx.pdf](#)

1. Triangle Cases According to the text, which two specific oblique triangle cases (represented by letter acronyms) are solved using the Law of Cosines?

2. Completing the Formula If you are solving for the square of side b , the Law of Cosines begins with $b^2 = a^2 + c^2$. What is the rest of the formula?

3. The Included Angle The general definition states that you subtract twice the product of two sides times the cosine of which specific angle?

4. The 90-Degree Simplification If the angle in the Law of Cosines happens to be exactly 90 degrees (meaning its cosine is 0), what famous theorem does the equation automatically simplify into?

5. Mental Semiperimeter Heron's Formula requires the "semiperimeter" (s). In Example 4, a triangle is given with sides $a = 1$, $b = 2$, and $c = 2$. What is the value of s for this triangle?

6. Mental Heron's Area Using Heron's Formula ($\sqrt{s(s-a)(s-b)(s-c)}$), what is the area of a right triangle with sides of 3, 4, and 5? (Hint: The semiperimeter is 6).

7. Defining Bearings Based on the text's definition of navigation, a bearing is always given as an acute angle measured from which two possible cardinal directions?

8. Reverse Bearing Example 7 features a flight from Airport A to Airport B at a bearing of N 50° E. If the pilot wanted to fly directly back from Airport B to Airport A, what would the new bearing be?

9. The Angle Formula If you need to find angle C and rewrite the Law of Cosines as $\cos C = \frac{\text{Numerator}}{2ab}$, what is the exact algebraic expression for the numerator?

10. Mental Law of Cosines Imagine a triangle where side $a = 3$, side $b = 8$, and the included angle $C = 60^\circ$. Knowing that the cosine of 60° is exactly 0.5, what is the length of side c ?